HASSAN ASAD

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience building web applications and REST APIs using Python and Flask, with familiarity with Django, JavaScript, React.js, and SQL. Developed full-stack and AI-powered applications involving backend development, database integration, RESTful APIs, NLP, and machine learning techniques. Strong foundation in object-oriented programming, database design, and web development. Seeking an entry-level Python/Django or Backend Developer role to strengthen my backend development skills and gain practical industry experience.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript (Basic), C++, PHP (Basic)
Backend & APIs: Flask, Django, Django REST Framework, REST APIs, REST Architecture
Frontend: HTML, CSS, Bootstrap, Tailwind CSS, React.js (Basic), Next.js (Basic)
Databases: SQLite, MySQL, PostgreSQL
Machine Learning & NLP: scikit-learn, Sentence-Transformers, spaCy, Pandas (Basic), TF-IDF, Neural Embeddings
Core Concepts: Object-Oriented Programming (OOP), Database Design, RESTful Architecture, SRS
Tools: Git, GitHub, VS Code

PROJECTS

TaskFlow – Project & Task Management System
Backend Web Application | Currently Developing

- Developing a web application using Python and Django for managing projects, tasks, users, and task assignments.
- Building RESTful APIs using Django REST Framework (DRF) for user, project, and task management.
- Using Django ORM and PostgreSQL for relational data modeling and database operations.
- Implementing authentication and user authorization to manage access to application features.
- Developing basic frontend functionality using HTML, CSS, and JavaScript to interact with backend APIs.
- Practicing debugging, problem-solving, and clean coding practices throughout development.

SmartCourse – AI-Powered Course Recommendation System
Final Year Project | BS Software Engineering

- Developed a full-stack web application that recommends online courses using natural language queries.
- Built a Flask REST API backend with 6 endpoints to support course recommendation and application functionality.
- Integrated two recommendation approaches: TF-IDF and Neural Embeddings for keyword-based and semantic course matching.
- Implemented NLP preprocessing using spaCy and machine learning functionality using scikit-learn and Sentence-Transformers.
- Designed an interactive Bootstrap 5 dashboard featuring search history, saved courses, and side-by-side comparison of recommendation results.
- Used SQLite for storing application data, including user history and saved courses.

TECHNICAL TRAINING

T3 Stack
UET Lahore | March 2024 – June 2024
- Completed comprehensive web development training covering TypeScript, Tailwind CSS, HTML, Next.js, React.js, and Node.js.

Web Development & Digital Marketing
Brains College, Lahore | November 2022 – February 2023
- Completed foundational training in web development and digital marketing.
- Built responsive web pages, dynamic forms, and basic CMS functionality using HTML, CSS, JavaScript, Bootstrap, PHP, MySQL, and WordPress.
- Gained exposure to SEO, SEM, and social media marketing strategies.

CERTIFICATIONS

- Cisco Networking Academy — Programming Essentials in C++
- Cisco Networking Academy — CCNAv7: Introduction to Networks

EDUCATION

Bachelor of Science in Software Engineering
Virtual University of Pakistan | 2026

Intermediate in Computer Science
Peak Solutions College, Saddar Cantt Campus | 2018 – 2020